# Lists: a tool for your marks

## Programming · Level 2 · Lesson 4



# Today we will

- Keep marks in a list and find them by position.
- Use len, sum, max and min.
- Loop over a list to count and average.

# Remember

- Positions count from 0: marks[0] is the first.
- The average: sum(marks) / len(marks).
- for m in marks: visits every mark in turn.

# A human list

1. Warm-up (5 min): Five pupils stand in a row with mark cards.
2. Teach (10 min): Index from 0; len, sum, max and min.
3. Ask the list (10 min): The right pupil answers each question.
4. Compare (10 min): Groups average by hand; the phone does too.
5. Wrap-up (5 min): Why from 0? Exit ticket.

**Use made-up marks only. Never put real pupils' names or marks on a shared board or a shared phone.**

# Let’s discuss

1. Why do you think lists count from 0?
2. What would sorted(marks) tell you that max(marks) doesn’t?
3. Which list in our school would you like Python to summarise?

# Exit ticket

- Write a list of three marks and the line that prints their average.